

Saad Tahir

Mechanical Design & Analysis | FEA & Computational Mechanics | Product R&D

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Professional Profile

Mechanical Engineer (M.S., Solid Mechanics) translating advanced research and analysis into practical, industry-ready tools and workflows. Brings published research, part-time founder experience, and cross-functional engineering exposure across centrifugal fan systems and corrugated packaging. Skilled in FEA, CAD, experimental mechanics, automation, and product improvement. Open to remote engineering roles and research collaborations.

Technical Expertise

Finite Element Analysis	Abaqus, ANSYS, OpenSeesPy, Inventor Nastran; static, Riks, buckling, modal and vibration analyses
Solid Mechanics	Composite mechanics and experimental validation
Mechanical Design	Autodesk Inventor (iLogic), SolidWorks; bespoke fan and impeller design, manufacturing drawings
Software Development	Python, MATLAB, Excel VBA; machine learning, optimization
Project Execution	SAP Business ByDesign; BOMs, PMs, $\text{\text{BOMs}}$, MS Word, MS Excel, MS PowerPoint

Education

University of Michigan–Shanghai Jiao Tong University Joint Institute (UM–SJTU JI)

Shanghai, China

Masters in Mechanical Engineering, Solid Mechanics (CGPA: 3.46/4.00 $\approx$ 86%, A)

2024

- Thesis: Computationally predicting corrugated container strength with creasing damage via nonlinear orthotropic lattice models.
- Honors: Full Graduate Fellowship, UM–SJTU JI (2019–2021); GRE: 322/340 (Oct 2018); TOEFL iBT: 112/120 (Oct 2018).

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

Bachelors in Mechanical Engineering (CGPA: 3.74/4.00 $\approx$ 94%, A)

2019

- Research: Feasibility assessment of solar thermal technologies for power generation and water desalination.
- Honors: NESCOM Fellowship, 1 of 3 recipients in a cohort of 130 (2017); Dean's Honors List with four merit awards (2015–2019).

University of Cambridge International Examinations (CIE), Beaconhouse School System

Islamabad, Pakistan

GCE Advanced & Ordinary Levels (A-Levels: 4 A*, O-Levels: 7 A*, 1 A)

2015

- Honors: Perfect SAT Subject Test scores (2400); National Aptitude Test (top 0.1%); Intl. Kangaroo Math Contest (top 1% nationwide).

Work Experience

Greybeard Corporate Solutions / Halifax Fan UK

Islamabad, Pakistan

Design & Analysis Engineer | Full-time

Jan 2025 – Present

Mechanical Design & Automation:

- **Fan Design & Order Delivery:** Delivered **130+ centrifugal fan designs** supporting a **£1.2M+ order portfolio** across UK, USA and China operations; engineered bespoke fan arrangements and impeller configurations in Autodesk Inventor.
- **Manufacturing Support:** Released manufacturing drawings, BOMs, PMs and production documentation through SAP Business ByDesign; collaborate closely with in-house purchasing and production teams, as well as clients, to ensure smooth fan delivery and resolve any post-design-release and aftermarket failure issues.
- **Practical Automation:** Developed scripts and iLogic rules, and tested/debugged tools with feedback in live order processing; supported CAD automation in production workflows and automated compliance documentation using Excel VBA.

Structural Analysis & Product Development:

- **Product R&D:** Developed an extensive FEA-driven impeller vibration prediction method with the University of Huddersfield (UK) to guide design decisions.
 - **Technical Analysis:** Predicted dynamic response using pre-stressed modal analysis and Campbell diagrams (Ansys Mechanical).
 - **Validation:** Simulated stochastic material and geometric variations to quantify uncertainty; validated with on-site hammer testing.
 - **Testing Setup:** Designed a cam–follower setup to initiate controlled fatigue cracks for post-damage performance assessment.
- **FEA-Driven Design:** Resolved vibration failure modes identified during testing, preventing delivery delays and field failures.
 - **Full Fan FEA Analysis:** Conducted full fan assembly FEA to isolate the physical source of the observed vibration failures during QA.
 - **Design Optimization:** Proposed FEA-based modifications offering simple, cost-effective implementations to resolve failures.

OnyxLeaf

Remote

Founder | Part-time

Nov 2025 – Present

- **Founder and engineering leadership:** Lead the initiative and its mechanical/structural engineering, coordinating software and business teams developing a cloud SaaS platform for automated corrugated packaging analysis.
- **Simulation & ML Development:** Architect the FEA platform by implementing updated physics models using Python and open-source FEA solvers for data generation; lead integration of FEA-generated datasets and composite mechanics-based feature engineering to develop surrogate models for automated box-strength prediction, achieving 98% ML-FEA accuracy with rapid analysis times under 1 second [Video].
- **Product strategy:** Partner with business and software teams on technical product strategy and data-driven investor pitch decks.
- **Research and mentorship:** Collaborate with NUST faculty and mentor undergraduate and postgraduate engineers in composite mechanics and experimental validation.

University of Michigan–SJTU Joint Institute

Graduate Researcher

Shanghai, China

Sep 2019 – Dec 2024

- **Advanced Predictive Engineering:** Developed a lattice FEA model for box buckling performance prediction incorporating homogenized panels and failure limits; achieved ~90% strength accuracy and realistic damage patterns validated against Box Compression Tests [\[Video\]](#).
- **Computational FEA:** Automated multilevel lattice-based FEA of box structures across buckling and nonlinear Riks analyses.
- **Nonlinear Constitutive Modeling:** Developed an original orthotropic multilayer beam-lattice model to emulate sandwich structures; calibrated via strain-energy equivalence, achieving ~98% agreement with high-fidelity shell models for multi-layer configurations.
- **Damage Mechanics:** Modeled creasing damage using nodal stiffness degradation within the lattice framework; derived analytical failure-limit equations accounting for liner stress concentrations from local buckling and stiffness mismatch at liner-fluting joints.
- **Experimental Characterization:** Characterized packaging behavior via paper tensile, board four-point bending, box compression, and impact testing per ASTM D685/D790/D642; employed UTM with NI-DAQ, LabVIEW, and dataloggers for validation.

School of Mechanical & Manufacturing Engineering (SMME), NUST

Undergraduate Researcher

Islamabad, Pakistan

Jul 2017 – Jan 2019

- **Solar Thermal Feasibility:** Conducted techno-economic evaluations of Linear Fresnel CSP plants using the NREL System Advisor Model, providing data-driven insights for renewable energy generation across Pakistan. First-authored journal publication and presented poster at an international conference.
- **Desalination Research:** Processed experimental data for a solar-powered water desalination still focused on decentralized water purification for remote regions, contributing to write-up and co-authoring conference paper.

Mari Petroleum Company Limited

Operations Intern

Sindh, Pakistan

Jun 2018 – Jul 2018

- **Asset Integrity:** Assisted gas production activities, including flow monitoring and quality checks (UT thickness, cathodic protection).
- **ESG & Sustainability:** Reviewed alignment with UN SDGs and assessed community-focused stakeholder engagement practices.

Scientific and Engineering Services Directorate (SES)

Engineering Intern

Islamabad, Pakistan

Jul 2017 – Aug 2017

- **Process Quality:** Observed CNC machining and fabrication methods; trained on GMAW/GTAW welding defect recognition and thermal distortion control.
- **Technical Standards:** Trained in Non-Destructive Testing protocols (VT, PT, MT, ET, RT, UT) and helium leak testing for critical component verification.

Publications

- **Tahir, S.,** Johnson, S., et al. (2022). Creasing damage analysis in corrugated packages using beam lattice models with joints stiffness degradation. *ASME IMECE*. [\[Paper\]](#)
- **Tahir, S.,** Abd-ur-Rehman, H.M., et al. (2021). Techno-economic assessment of concentrated solar thermal power generation and potential barriers in Pakistan. *Journal of Cleaner Production*. [\[Paper\]](#)
- Abd-ur-Rehman, H.M., Shakir, S., Saqib, H., & **Tahir, S.** (2018). Decentralized and cost-effective solar water purification system. *IOP Conference Series: Earth and Environmental Science*. [\[Paper\]](#)

Selected Engineering Projects

- **Advanced Mechanics of Composite Materials:** Built a controlled UV-curing setup for single-glass-roving composites and performed ASTM D790 three-point bending (MTS); processed load-deflection data to extract modulus trends (MATLAB).
- **Engineering Optimization:** Implemented an analytical model of a bioinspired tunable-stiffness mechanism and optimized geometry/layout using `fmincon` and multi-objective methods (ϵ -constraint, NSGA-II) (MATLAB).
- **Advanced Measurement & Instrumentation:** Processed Pressure-Sensitive Paint (PSP) wind-tunnel images to recover surface pressure fields using image registration, Stern-Volmer calibration, and filtering (MATLAB).
- **Data Mining:** Built a defect-classification pipeline for magnetic-tile images including feature extraction (PCA/SVD, HOG) and benchmarking of kNN, SVM, and CNN (MATLAB).
- **Capstone (Bachelors):** Designed, manufactured, and validated a hydrokinetic micro-turbine for low-head, low-flow canals.

Volunteer

NUST Society of Mechanical Engineers (NSME)

President (formerly Publications Chair)

NUST, Islamabad

Mar 2017 – Mar 2019

- Supervised 40+ members and spearheaded training workshops, mock recruitment drives, and entrepreneurial competitions; pioneered a [blog](#) for the society's project resource catalog.

Solar Decathlon Middle East 2018

Project Coordinator

NUST, Islamabad

Sep 2017 – Feb 2018

- Managed [Team BGreen](#), the first Pakistani team selected for the competition, coordinating 20+ architecture and engineering graduates to deliver on schedule.